

premium models—are now cascading into mass-market vehicles. Panoramic dashboard-spanning displays, curved organic light-emitting diode (OLED) panels, and augmented-reality HUDs are setting new benchmarks for automotive user interface and user experience (UI/UX).

In India, this evolution is gaining significant traction. As the automotive market pivots toward electric and connected mobility, consumers are increasingly demanding technology-rich, safe, and personalised driving experiences, driven by digital interfaces. Even in two-wheelers, commercial fleets, and utility vehicles, ruggedised displays are being embedded for diagnostics, telematics, and navigation.

With display content per vehicle rising rapidly, and the automotive display market expected to surpass \$12 billion globally by 2027, the opportunity for India is two-fold:

- To meet rising domestic demand with localised, cost-effective, automotive-grade displays
- To participate in the global supply chain as a competitive source of modules, sub-assemblies, and integrated display-HMI solutions

Yet, the path is not without its challenges. Stringent safety standards, complex integration with vehicle electronics, and durability requirements set a high bar for quality and reliability. Seizing this opportunity requires strategic alignment between industry capabilities, policy support, and focused research and development (R&D)—a theme that emerged prominently during the Society for Information Display (SID) India workshop in June.

India's auto industry: The rise of displays

India has quietly emerged as the world's third-largest automobile market, with over four million passenger vehicles sold annually and a vast two-wheeler and commercial vehicle base. Yet, until recently, the interior digital experience remained

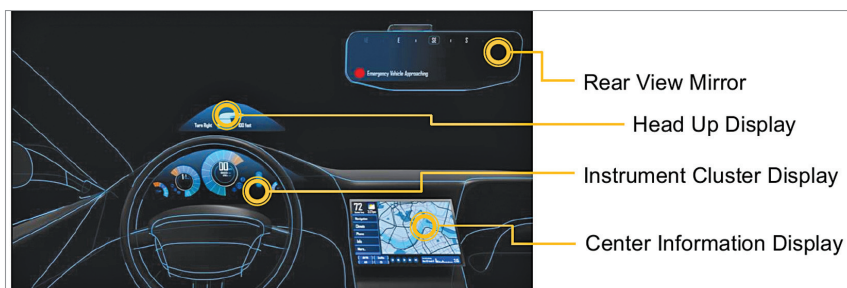


Fig. 1: An overview of different automotive displays

relatively underdeveloped, limited largely to basic infotainment in top-end variants.

This is now changing rapidly. As electrification, connectivity, and shared mobility reshape consumer expectations, Indian automotive original equipment manufacturers (OEMs) are responding with smart, screen-rich cockpits even in mid-segment models. Electric vehicles from Tata, Mahindra, and emerging players such as Ola Electric are integrating larger, touch-enabled displays that combine navigation, media, diagnostics, and telematics. The display is no longer a luxury; it is a functional necessity and a branding differentiator.

Moreover, the Indian buyer, digitally savvy and increasingly aspirational, is demanding global-grade cockpit experiences.

However, while demand is surging, most automotive displays continue to be imported or integrated via global Tier-1 suppliers, adding cost and limiting supply chain control. Indian EMS providers have yet to scale to automotive-grade certifications, and very few domestic firms are designing display systems from the ground up.

EMS and display manufacturing

India's electronics manufacturing sector has made remarkable progress over the past decade, evolving from assembly-driven units to vertically integrated EMS providers catering to mobile phones, consumer electronics, and industrial applications. However, when it comes to automotive-grade displays,

this momentum is yet to fully translate.

The opportunity is substantial. Automotive displays require not just screen panels but advanced integration capabilities, including optical bonding, ruggedised enclosures, electromagnetic interference (EMI) shielding, embedded controllers, and rigorous reliability testing. Indian EMS players such as Dixon, Syrma SGS, VVDN, and Bharat FIH have begun moving up the value chain, but certified Tier-1-level display module manufacturing remains nascent.

Most display panels—including thin-film transistor liquid crystal display (TFT LCD) and organic light-emitting diode (OLED) technologies—continue to be imported, primarily from China, Korea, and Taiwan. Key elements such as backplane technology, cover glass, bonding materials, driver integrated circuits (ICs), and touch sensor integration are not yet fully localised. The automotive sector further demands long product lifecycles, temperature resilience, vibration resistance, and compliance with safety standards such as AEC-Q100, ISO 26262, and Automotive Safety Integrity Level (ASIL) requirements, raising the bar for market entry significantly.

Yet, several green shoots are visible:

- Select EMS firms have begun investing in optical bonding lines and cleanroom facilities tailored to automotive assemblies.
- The display research ecosystem is growing through the display labs at IIT Kanpur and IIT Chennai.
- Academic interest in display physics, optics, and design is